

RESULTS

Results

01 · PRINCIPAL COMPONENT ANALYSIS (PCA)

reduce variables to components

Table 1. Variance explained

Component	Eigenvalue	% variance	Cumulative %
PC1	3.216	35.7	35.7
PC2	1.639	18.2	53.9
PC3	1.365	15.2	69.1

Table 2. Component loadings

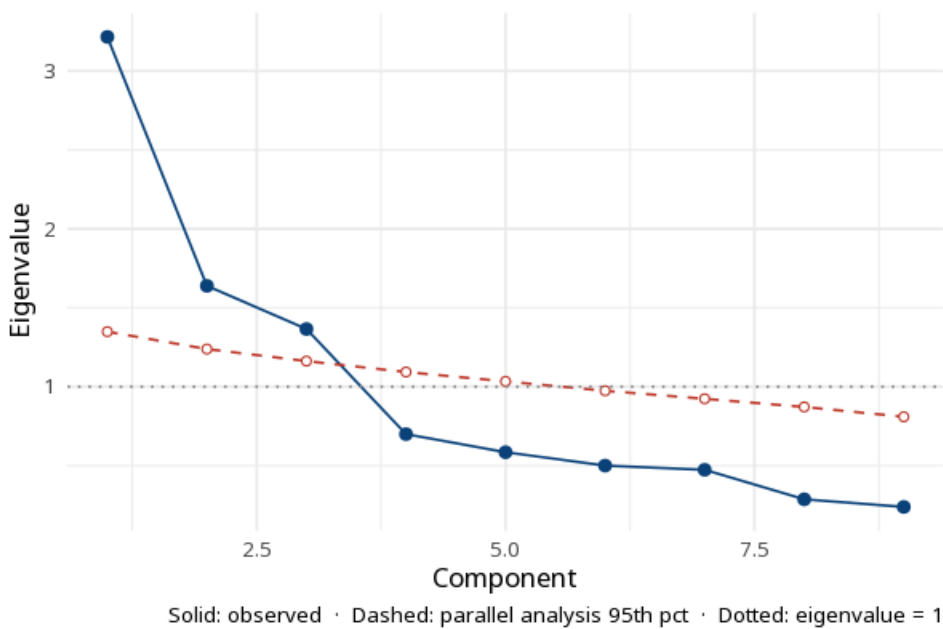
Variable	PC1	PC2	PC3
x1	.66		−.37
x2	.39		−.62
x3	.48	.33	−.54
x4	.77	−.45	
x5	.74	−.48	
x6	.77	−.43	
x7	.35	.50	.59
x8	.45	.61	.33
x9	.59	.51	

Scope: PCA is data reduction — components are weighted composites, not latent factors; communalities are not reported (Jolliffe & Cadima, 2016; Frick et al., 2025).

Method: Component loading columns expand to the number of retained components; loadings are correlation-scaled (eigenvector × $\sqrt{\text{eigenvalue}}$).

Cutoffs: Loadings $|< .32|$ are suppressed.

Figure. Component retention



Understanding the numbers

Component. One retained principal component - a weighted composite of the observed variables, not a latent factor. Here, 3 component(s) were retained (rows below), ordered by variance explained.

Eigenvalue. How much variance a component retains; components are retained where the observed eigenvalue exceeds the parallel-analysis threshold (the Kaiser > 1 rule is shown for reference but tends to over-extract, Zwick & Velicer, 1986). Each retained component's eigenvalue already cleared the retention rule you selected.

% variance. The share of total variable variance one component retains on its own. Each row's % variance is that component's own contribution.

Cumulative %. The running total of variance retained as components accumulate; $\geq 70\%$ is a common but explicitly subjective target (Jolliffe & Cadima, 2016). Here, the retained components together explain 69.1% of total variance.

Variable. One observed variable contributing to the components. Each row is one observed variable's own set of component loadings.

PC1. How strongly (and in which direction) a variable contributes to PC1; loadings are correlation-scaled and values below $|.32|$ are left blank. A blank cell here means that variable's loading on PC1 fell below the $|.32|$ display threshold.

PC2. How strongly (and in which direction) a variable contributes to PC2, on the same correlation-scaled $|.32|$ rule as PC1. A blank cell here means that variable's loading on PC2 fell below the $|.32|$ display threshold.

PC3. How strongly (and in which direction) a variable contributes to PC3, on the same correlation-scaled $|.32|$ rule as PC1/PC2. A blank cell here means that variable's loading on PC3 fell below the $|.32|$ display threshold.

How to read this test

PCA compresses correlated variables into a few components. The variance explained (Table 1) shows how much information each component retains; the component loadings (Table 2) show how strongly each variable contributes.

Loadings are correlation-scaled (eigenvector $\times \sqrt{\text{eigenvalue}}$) so they share the unit of a correlation coefficient. PCA is data reduction — components are weighted composites of the observed variables, not reflective latent factors; do not interpret them as constructs and do not report communalities (which belong to factor analysis, not PCA). For a latent-construct model use EFA (psych::fa) or CFA (lavaan). The scree plot overlays the parallel-analysis 95th-percentile threshold — retain components where the observed eigenvalue exceeds that line; the Kaiser eigenvalue > 1 rule is shown for reference but tends to over-extract (Zwick & Velicer, 1986). Cumulative variance $\geq 70\%$ is a common but explicitly subjective target (Jolliffe & Cadima, 2016).

APA template: PCA (correlation matrix; parallel analysis) retained 3 components explaining 69.1% of total variance.

Statistical basis: Principal component analysis as a data-reduction method (not a latent-variable model) (Jolliffe, I. T., Cadima, J. (2016). "Principal component analysis: a review and recent developments." Philosophical Transactions of the Royal Society A, 374, 20150202.); PCA is not EFA - components are composites, not reflective factors (Frick, S., Killisch, R., Bißantz, N., Wetzel, E. (2025). "PCA Is Not EFA." European Journal of Psychological Assessment, 41(6), 425-426.); Kaiser eigenvalue > 1 rule tends to over-extract (Zwick, W. R., Velicer, W. F. (1986). "Comparison of five rules for determining the number of components to retain." Psychological Bulletin, 99(3), 432-442.). Full references in the exported CITATIONS.txt.

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